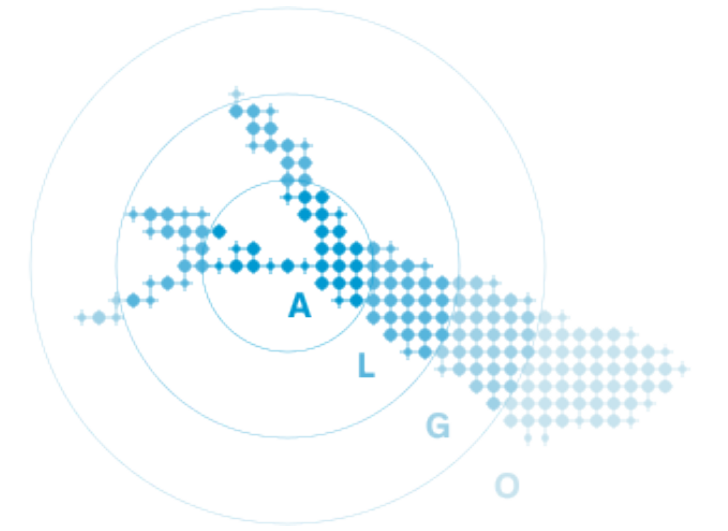




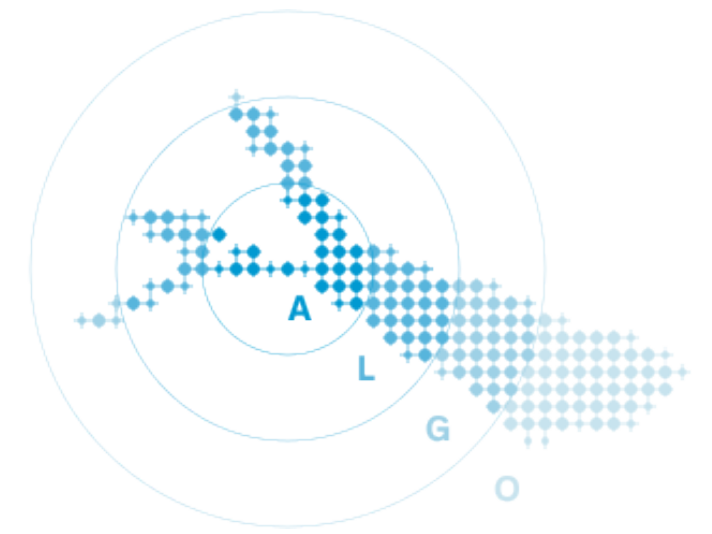
Apr 2025

An Interactive and Introductory IPE Tutorial

Basics of IPE & Creating Presentations

Algorithmics Chair of the University of Konstanz

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PgDown



Apr 2025

An Interactive and Introductory IPE Tutorial

Basics of IPE & Creating Presentations

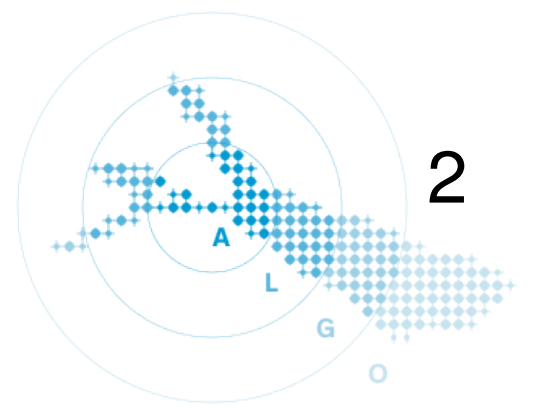
Algorithmics Chair of the University of Konstanz

Action instructions are in grey,
shortcuts will be shown
in **seegrün**

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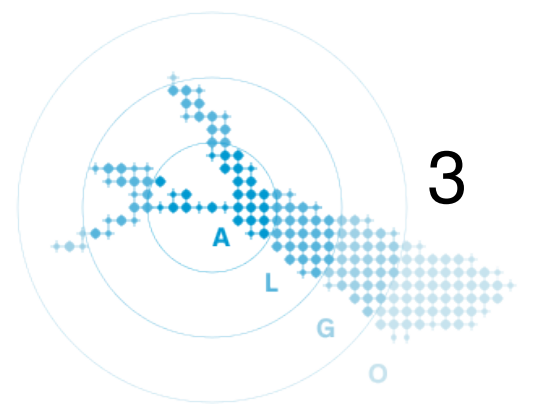
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What is IPE and why you should use it!

To go to the previous page, press
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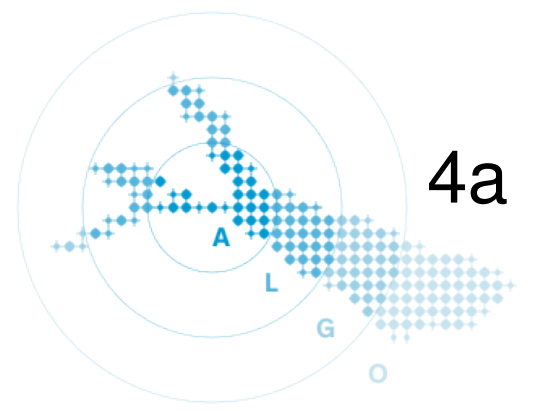
What is IPE?

- At its core, IPE is a drawing editor
- It can be used to create professional figures (e.g. for a thesis)
- Presentations like this one (or for your seminar/colloquium...or Prof. Storandt's lecture slides) can easily be created as well
- IPE is a tool specifically aimed at the scientific community

Annoyed by the grid and green boxes?

Press **F12** to toggle the grid on/off

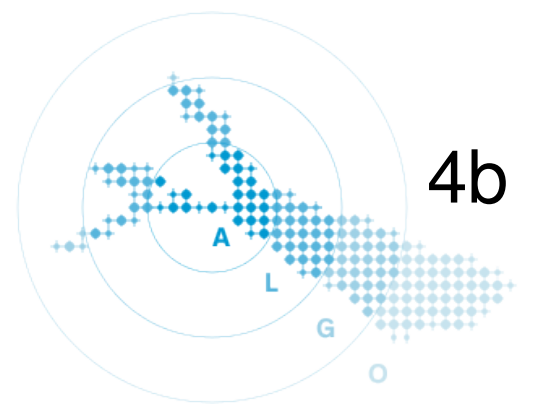
Ctrl+F12 to toggle the bounding boxes



What makes IPE so useful

- IPE not only features useful drawing tools and snapping modes, but also has full $\text{\LaTeX}$ support!
- Any text or label you add to your figures and presentations is interpreted as $\text{\LaTeX}$ text
- This means that, for example, adding equations like $\sum_{i=1}^n n^2 = ?$ (I forgot the formula) is easy
- Same goes for pseudocode:

```
for  $i = 1 \dots n$  do  
   $\text{\textit{sum}} = \text{\textit{sum}} + i^2$   
return  $\text{\textit{sum}}$ 
```



What makes IPE so useful

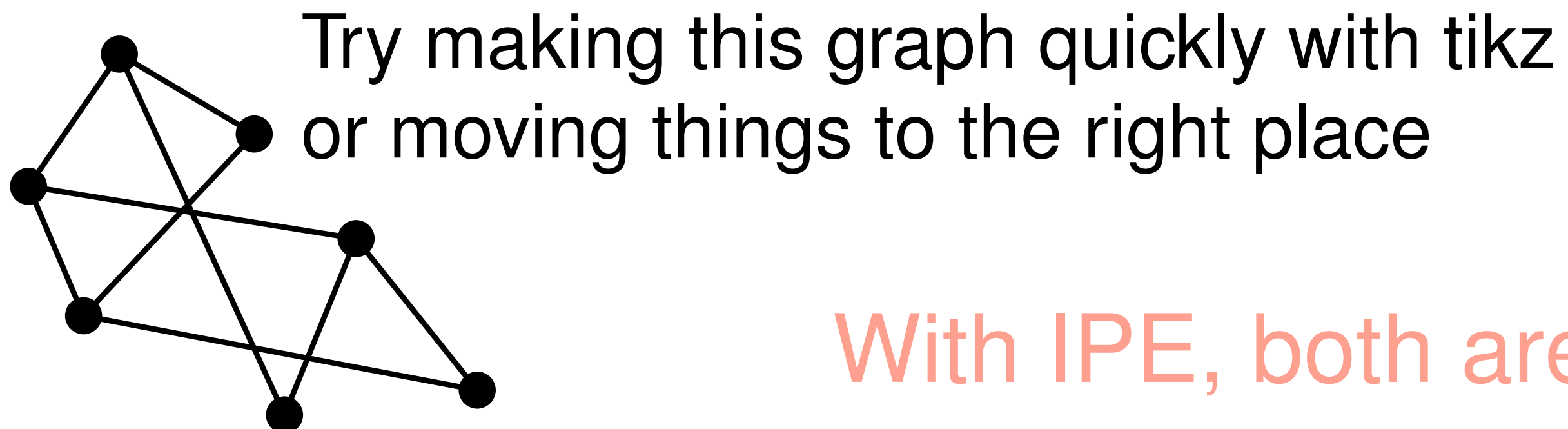
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- Same goes for pseudocode:

```
for  $i = 1 \dots n$  do  
   $sum = sum + i^2$   
return  $sum$ 
```

Check it out yourself: Select (**S** for selection mode) the list with the **left mouse**, go to the *context menu* with the **right mouse** and select *Edit text*. See the $\text{\LaTeX}$ commands?

What makes IPE so useful

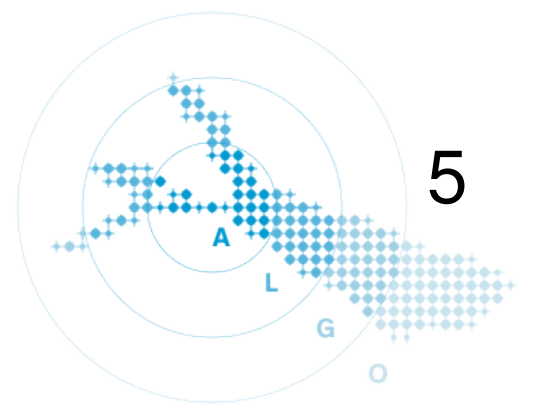
- Basically, with IPE you get the power for scientific notation and tools provided by $\text{\LaTeX}$
- But at the same time, also the ease of visual arrangement and tuning that drawing programs offer
- For presentations, IPE is sort of a sweetspot between Latex Beamer and Powerpoint



Or adding pseudocode or
equations in PowerPoint

With IPE, both are easy!

```
for  $i = 1 \dots n$  do
   $\lfloor \text{sum} = \text{sum} + i^2$ 
return  $\text{sum}$ 
```



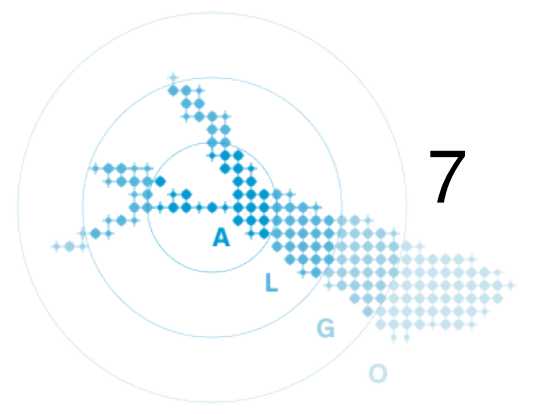
Other cool properties of IPE

- IPE is super extensible and configureable:
- New functionality can be added by writing Lua scripts
- Style sheets allow defining colors, symbols, background templates and many more, and can be easily modified
- IPE produces PDF documents and any PDF that was created by IPE can be opened in it and modified (like this tutorial)
- You can steal other people's style sheets
- IPE is actually quite intuitive and easy to use
- Many shortcuts for fast editing

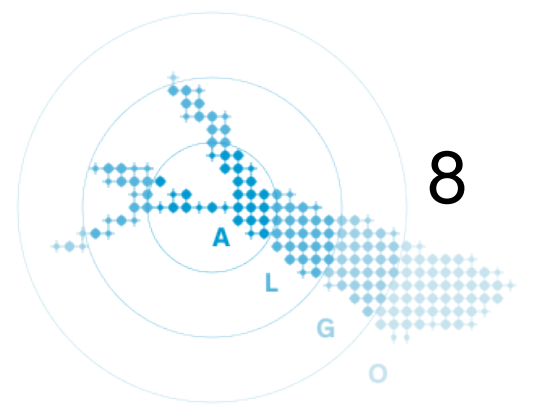
Compilation

- IPE needs to compile $\text{\LaTeX}$. There are two options:
- You have the option to let IPE compile online which requires no local $\text{\LaTeX}$ distribution. Internet connection is obviously required and your text will be forwarded to the compilation service.
- If you have a $\text{\LaTeX}$ distribution installed on your machine, it can be used. This can be helpful if you want to use IPE in places where you don't have internet connection. You need to ensure that your installation has all packages used by the file you are editing.

To switch between these options, go to the *Help* tab and select *Enable online Latex-compilation*. In the dialogue you can either enable or disable it, depending on what you are currently using.



Basics of using IPE

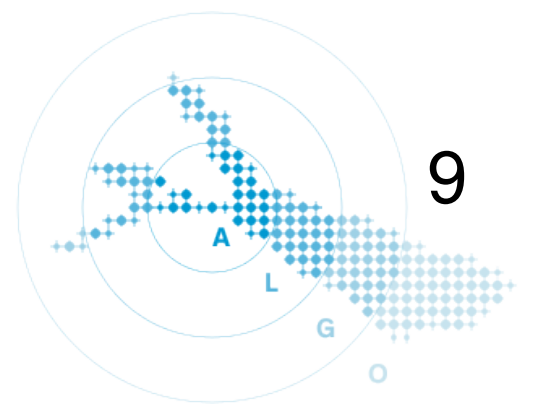


Moving around

- Using **Ctrl+Zoom** (Mousewheel/Touchpad) is the most standard way to get your page where and how close you need it
- The cursor is used as a reference point for zooming into/out of, so e.g. zooming in to the left of the page will make the page go to the right (you'll get it, it's similar to other drawing programs)
- If you're lost, pressing **x** centers on the location of the cursor by panning
- If you're totally lost, pressing **** (backslash) fits the page to your view

Try it! Zoom out with the cursor to the left of the page, then center on it by pressing x and then zoom in again.

Selecting



- Selecting in IPE can be a bit tricky, but first go to Selection mode by pressing **S**
- Objects are selected with the **Left Mouse**
- Important: IPE only considers an object's border for selecting which is why enabling the green boxes (**Ctrl+F12**) is helpful

Try selecting this rectangle by clicking in the middle of it. Nothing happens. Now try selecting by clicking on or close to the pink border. Works.

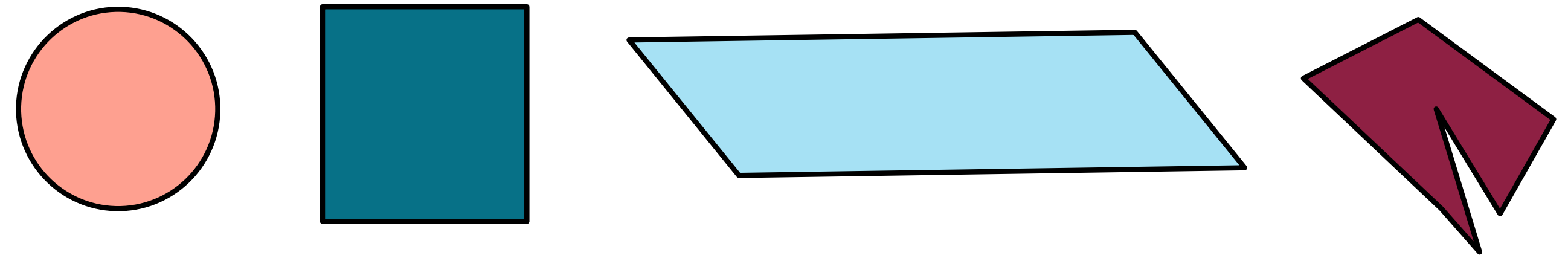


- Deselecting works by clicking somewhere where there is no object (e.g. off the page) or by selecting the same object again while holding down **Shift** (Now deselect the rectangle)

Further selection tricks

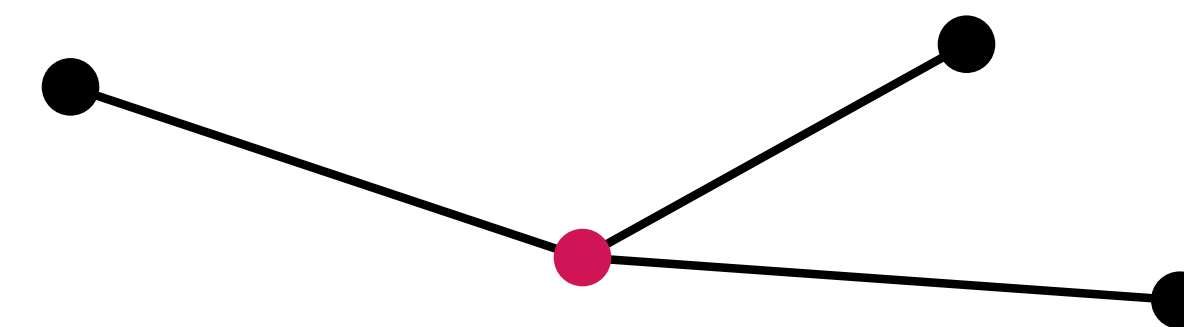
- As common in other programs, while holding down **Shift**, you can select multiple objects

Here are some practice objects:



- You'll notice one object of a selected group (last selected) has a different outline. This is the reference object for aligning etc.
- If multiple objects are close, selecting the right one may require **holding down the left mouse** and tabbing through them with **Space**

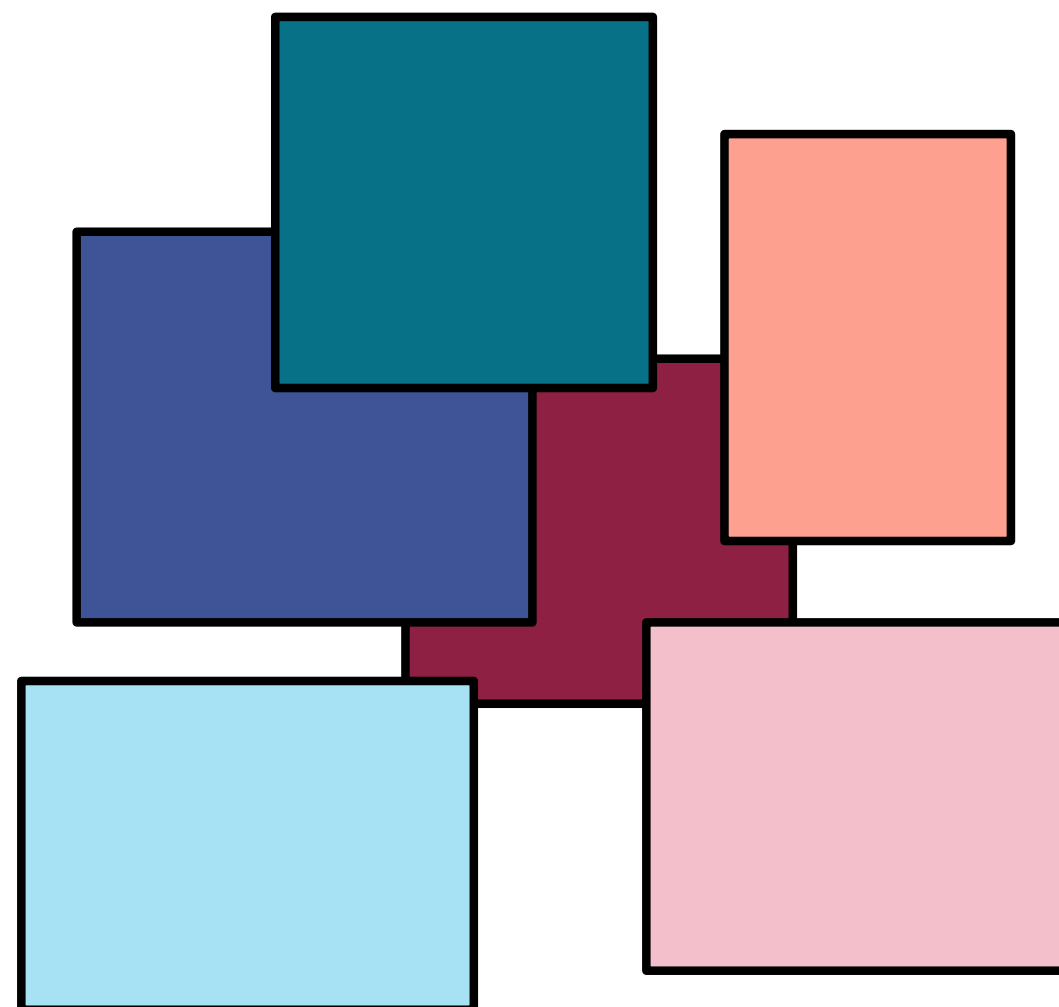
Try selecting the red vertex in this graph:



Further selection tricks

- You can also select (multiple) objects by making a box through holding down the left mouse button and moving it
- If you go from left to right, it selects all objects entirely contained in the box
- But if you go from right to left, it selects all objects that intersect the box

Try selecting only the blue-green-ish rectangles on the left using these techniques



Or only the dark blue plus both red-ish ones (forming the diagonal from top left to bottom right)

Drawing shapes

- You can draw a bunch of different shapes using the tools in the top bar
- Shapes are each defined by a characteristic set of control points which are placed with the **Left Mouse**
- **Shift** may act as a modifier (e.g. for making squares with the rectangle tool)
- The control points of lines can be undone by pressing **Del** and lines are finished using a **double-click**

Draw your favorite shape here:



Drawing shapes

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Draw your favorite shape here:
Now delete it, by selecting
it and pressing **Del!!**



Drawing shapes

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Draw your favorite shape here:

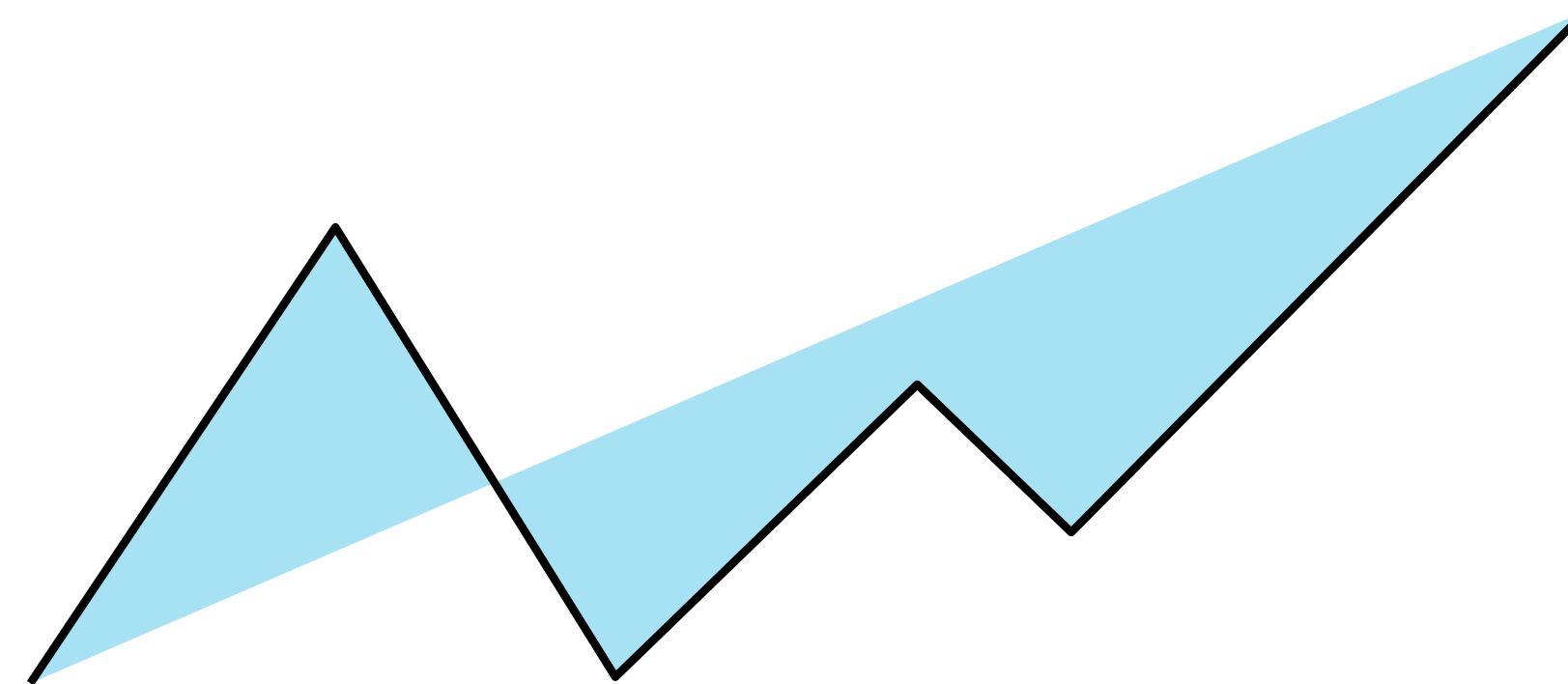
(This is how you delete anything) → Now delete it, by selecting it and pressing **Del!!**



Object properties

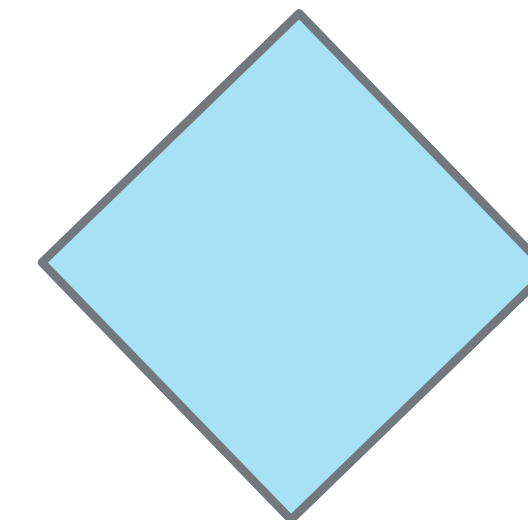
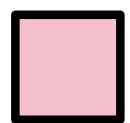
- Your new object gets the properties that are currently displayed on the left
- To change your object, either select it and then change the properties on the left or go to the context menu with **Right Mouse** and change it there

Change this object's properties to make it a fat green line (no fill) with a large forward arrow



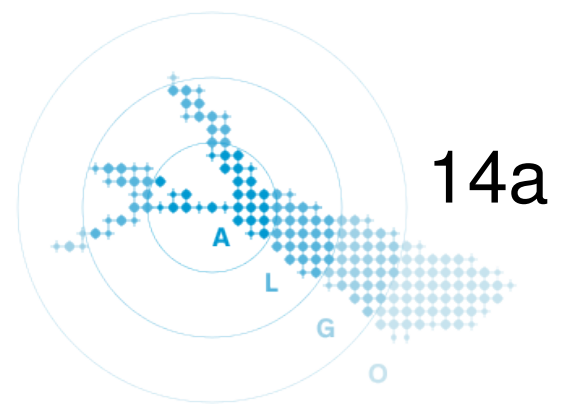
Translate, rotate, stretch

- Move objects around by going to Translating mode by pressing **T**
- Rotate objects in Rotating mode: **R**
- Make objects bigger/smaller/stretch them in Stretching mode: **E**
- **Shift** acts as a modifier:
- When translating, it allows moving either horizontally or vertically, but not both
- When stretching, it actually scales the object, keeping its proportions



Make the pink square over there cover this blue one

Snapping

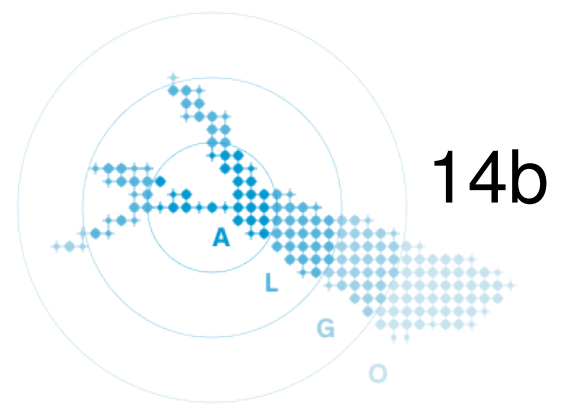


- IPE offers a couple of snapping modes that are very useful for making nice-looking figures. They can be toggled at the very top left or via shortcuts
- Use grid snapping (F7) to aid making regular alignments
- You can change the grid size in the middle of the top toolbar (where it says 'xy' pts)

Use grid snapping to make a new square of equal size to the green one, spaced exactly one such square size apart to the right (you may have to change grid size)

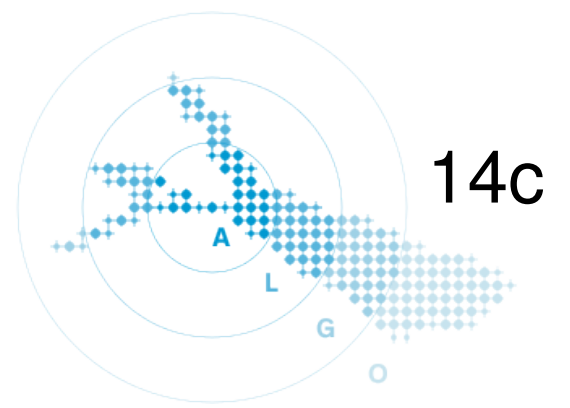


Snapping

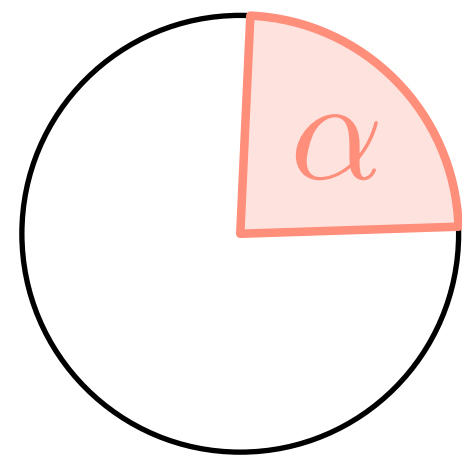


- Vertex snapping (**F4**) allows snapping to end points or corners of shapes
- Control point snapping (**Shift + F4**) allows snapping to the points that were used to make a shape
- Boundary snapping (**F5**) allows snapping to continuous boundarys (edges) of objects or the page's frame

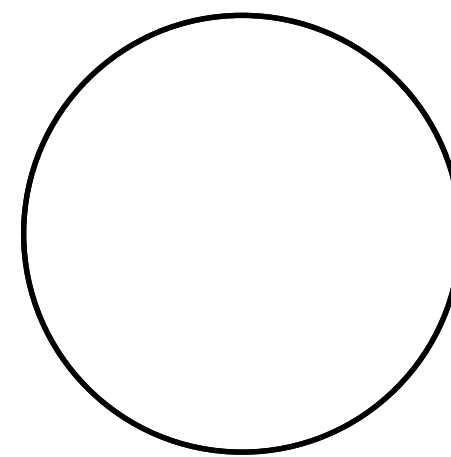
Snapping



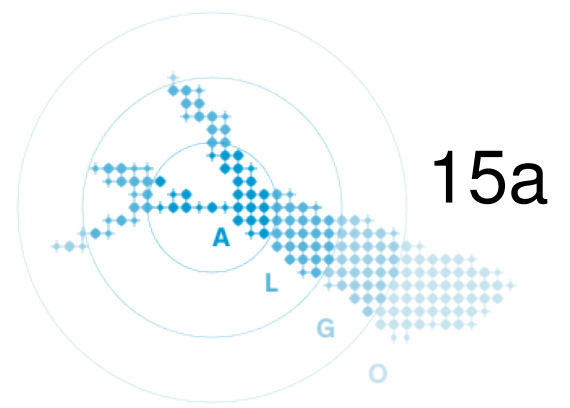
– Let's use these snapping modes to make a cool figure!



- I. Select your favorite color in the top color property field (stroke color)
- II. Trace parts of the arc of the circle at the bottom with a circular arc (**A**).
Select the circle center first (control point snapping), then two points on the boundary (boundary snapping)
- III. Now create a line (**P**) that goes from one end of the arc you just created (vertex snapping) to the circle's center and then to the arc's other end
- IV. Select both, the arc and the wedge shaped line, right click and then select *Join paths*
- V. Give your new shape a fill (if it doesn't have one already) and your result should look something like the one on the left (label optional)



Text objects



- There are multiple ways to add text to your figures/presentations:
- A text label (L) is the simplest, creating a text object that fits to its content's width (typically one line)
- For multiple lines in a label, change its *Type* to 'minipage' in the context menu and use manual line breaks
- For a paragraph (L), you can manually define its width and text will be broken into lines automatically according to the width
- Changing the width is possible through the context menu
- Mainly for presentations, a text box (F10) inserts a text object that spans the entire page frame below the lowest text on the page

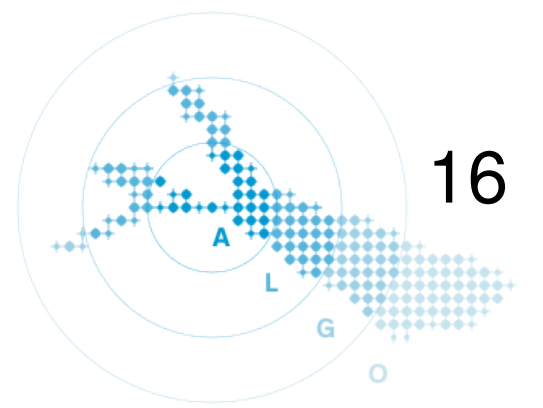
Text objects

- All text objects have two options:
- The *Style* defines the environment, e.g. math, itemize etc.
- The *Size*...well, you can guess

Feel free to try creating labels,
paragraphs or text boxes here

If you want to rotate a label or
paragraph, you need to change its
Transformation to 'rigid' first
through the context menu

Aligning



- Options for aligning and distributing objects can be found under *Ipelets > Align & distribute*
- Objects are moved to align with the reference object
- Logical shortcuts for aligning use the modifier **Shift** and:

T for aligning at the top

B for aligning at the bottom

L for aligning on the left side

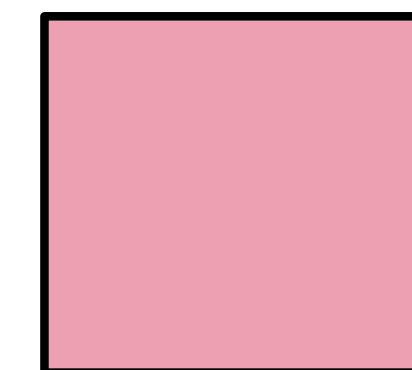
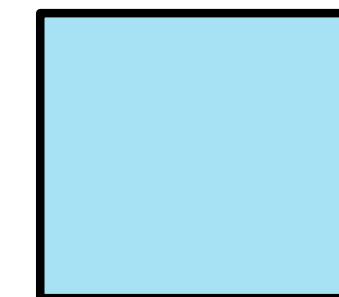
R for aligning on the right side

C for aligning the center points

V for aligning vertical center points

H for aligning horizontal center points

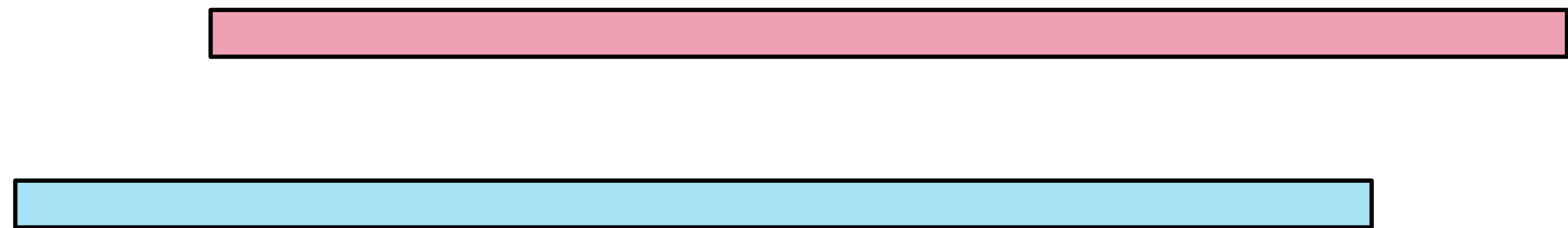
Here are two objects
to try it out on:



Distributing and other alignment tricks

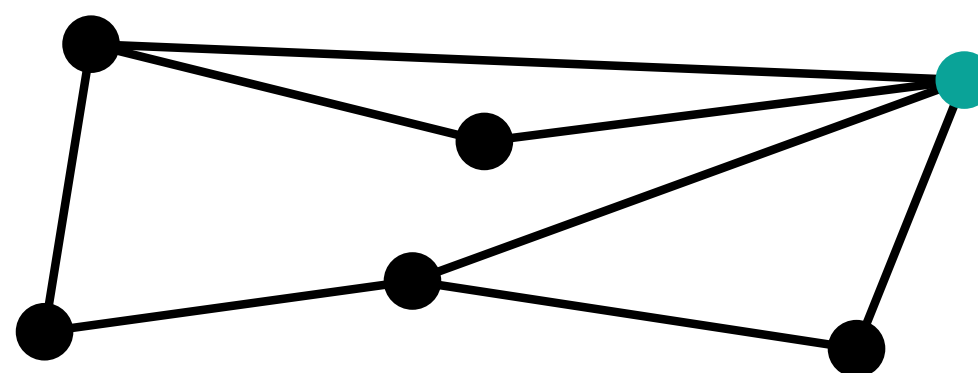
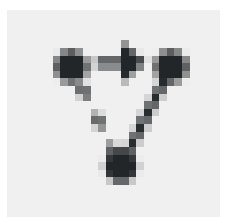
- Distributing multiple objects is possible from the same menu as aligning (no predefined shortcuts)
- The option to *distribute top to bottom* can be interesting to match one object's top to another objects bottom, i.e. putting them right above another
- If you *set skip* (a value in pts) first, that amount of space will be left between the objects

Make these two bars
align horizontally and
space them 10 pts
apart vertically



Drawing graphs

- As computer scientists (especially algorithmics enthusiasts), we need to draw graphs from time to time. IPE makes this easy!
- Use marks (**M**) as nodes/vertices. Mark shapes are configurable through the properties (third from the bottom), but ‘disk’ works well
- Enable vertex snapping mode and draw each edge as a separate line (**P**) between two vertices (should show a red guide element)
- If you do this, you can now move your graph nodes with the corresponding tool, and all connected edges will follow!



Move the green node with this tool and see the effect!

Draw your own graph here:



Style sheets

- IPE has style sheets. Think of them like a CSS file. Technically, they're XML.
- Primarily, the style sheets define all of the property options you have for your objects (e.g. colors, line widths, text sizes etc.)
- They can also control the appearance of your page(s), including possible templates like the one used in this presentation
- They can define a set of $\text{\LaTeX}$ packages you always need
- Basically, style sheets can do a lot and are advanced IPE stuff
- If you want to check or edit your/this/any document's style sheets, press **Ctrl+Shift+S** for options (more on how style sheets work in the official IPE documentation)
- **Ctrl+Shift+U** updates them if you made and saved changes

Making presentations in IPE

(This is what you're here for anyway, right?)

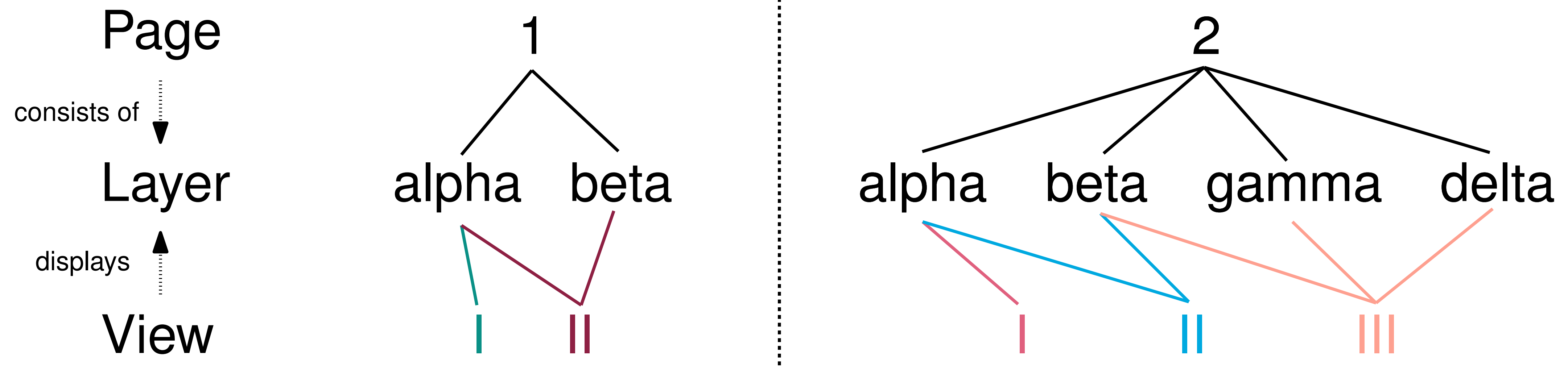
(But don't sleep on IPE's potential to make figures for your thesis)

Presentation concepts: Page, layer, view

- A presentation made with IPE is simply a set of PDF slides. This means no fancy transitions, just plain content.
- IPE lets you create *pages*, *views*, and *layers*. Here's how they translate to your final presentation:
- Each *page* in IPE is a new piece of canvas for you to draw on.
- Each *view* in IPE will be a new slide in your final PDF. Each IPE page may have multiple views.
- IPE's *layers* allow you to group objects and manage their visibility. Whether a layer appears on a slide is determined by whether it is part of the corresponding view. New objects get added to the *active layer* (yellow highlighting).

Presentation concepts: Page, layer, view

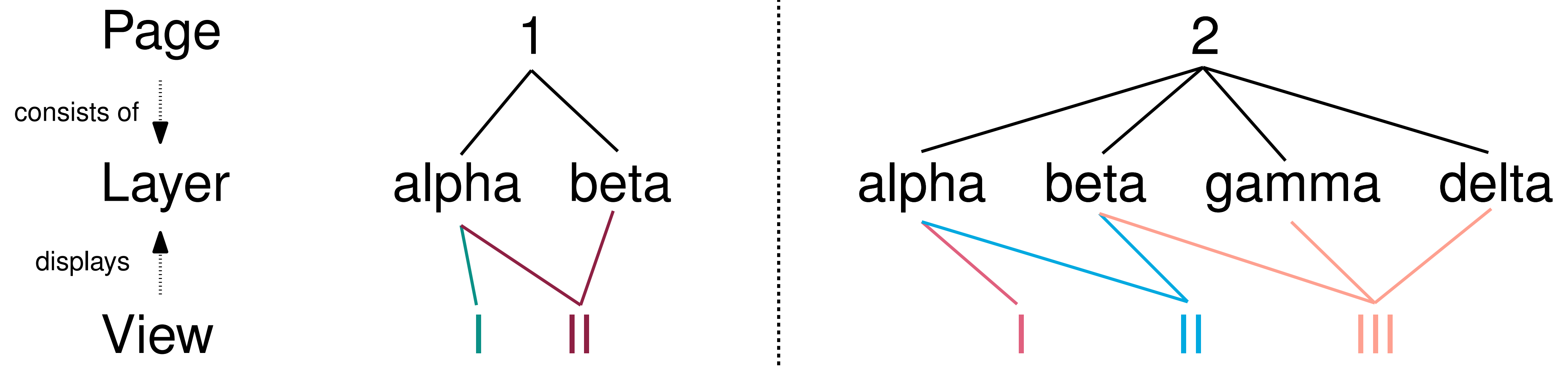
- Consider this structure for example:



- This final presentation will consist of 5 slides – one for each view.
- The first slide will show layer alpha of page 1. The second slide will additionally show the layer beta of page 1 and keeps alpha visible. This is typical for incrementally revealing e.g. bullet points.

Presentation concepts: Page, layer, view

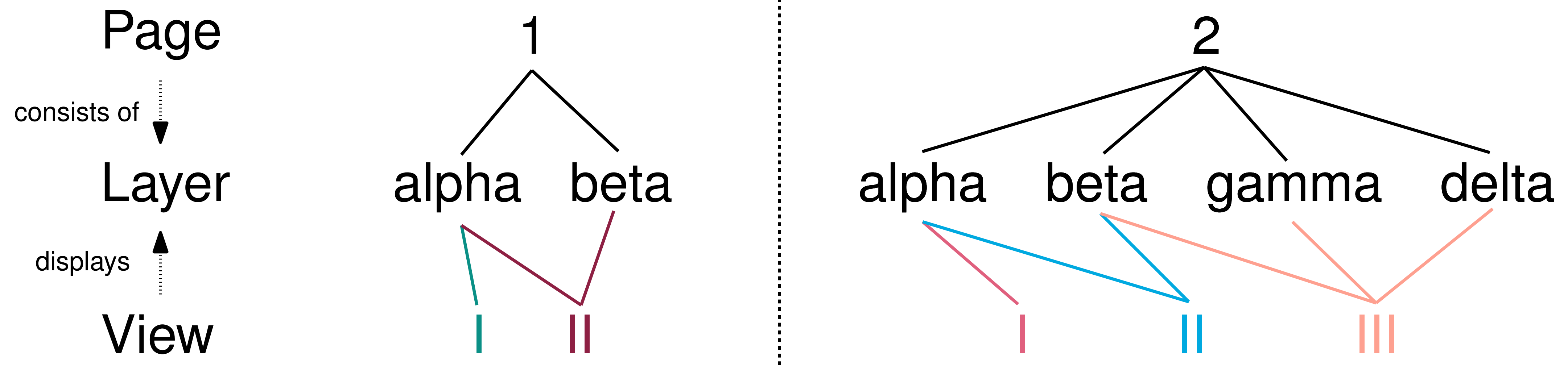
- Consider this structure for example:



- Notice how the last slide (view III of page 2) adds two new layers (gamma and delta) compared the view before and also drops the first layer? Everything is possible!

Presentation concepts: Page, layer, view

- Consider this structure for example:



Take a look at the layers and views of this page (shown on the left)!
 Play with which layers are active in which view to get a better understanding.
 If you enable all layers, you'll see that this IPE page is actually massively overloaded. This is not good practice (better to create a new page than overloading views and layers), but sometimes helpful.

Shortcuts

- New pages can be created with **Ctrl+I**
- Cutting/Copying/Pasting pages can be done with **Ctrl+Shift+X/C/V**
- A new layer can be created with **Ctrl+Shift+N**
- Most commonly, you will probably need the shortcut **Ctrl+Shift+I** to create a new view and new layer that gets activated for the new view
Try adding a new view and layer, add something to it and go back and forth to see the effect
- Deleting is not possible via shortcuts. Pages and views can be deleted from the corresponding tabs at the top, layers from the list on the left

Presentation style sheets

- Presentations require certain properties (landscape format, bigger font sizes etc.) that need a dedicated style sheet
- IPE's installation supplies a 'presentation' style sheet for 4:3 format slides in the subfolder 'styles'. Add this and remove the style sheet 'basic' from a new IPE project to get started with a presentation.
- Presentation style sheets can also define a design template for the pages

Using this template

- This template (contained in the ‘algoKN’ style file) gives you the look you see in this presentation (and for example all corporate design colors of the University of Konstanz)
- Here are some pointers on how to use it:
- To add a page title, press **Ctrl+P**. Here you can also create bookmarks for your PDF with the sections (pretty cool, right?)
- The bottom bar requires you to define the following commands in the Latex preamble of the document properties (**Ctrl+Shift+P**):
`\bottomleft \bottommiddle \bottomright`

Try to change the bottom bar and the title of this page now!

Using this template

- This template (contained in the ‘algoKN’ style file) gives you the look you see in this presentation (and for example all corporate design colors of the University of Konstanz)
- Here are some pointers on how to use it:
- The title page can be inserted by using a *symbol* via the tab *lpelets* > *Symbols* > *use symbol* and selecting ‘titlePage’ from the second drop down. Use vertex snapping to fit it on a page.
- It requires you to define the following commands in the Latex preamble of the document properties (**Ctrl+Shift+P**):
`\fulltitle \subtitle \name \subject \displayDate`
They may be left empty, but need to be defined.

Using this template

- This template (contained in the ‘algoKN’ style file) gives you the look you see in this presentation (and for example all corporate design colors of the University of Konstanz)
- Here are some pointers on how to use it:
- Page numbers can be (de)activated in the document properties and their appearance is determined by the style sheet
- `\ul{...}`, `\subheader{...}`, `\define{...}`
are (pre)defined commands set up to help you create a consistent appearance throughout this template

Try adding a label or text box here and use the define or subheader command:

Using this template

- This template (contained in the ‘algoKN’ style file) gives you the look you see in this presentation (and for example all corporate design colors of the University of Konstanz)
- Here are some pointers on how to use it:
- If you want to use this template, you are probably best off using the .ipe file that has this configured already
- Alternatively, you can copy the preamble from this document or from the Git repository

Using this template

- This template (contained in the ‘algoKN’ style file) gives you the look you see in this presentation (and for example all corporate design colors of the University of Konstanz)
- Here are some pointers on how to use it:
- Pro tip: If you do not want the template background (logo, bottom bar) on a given page, add a layer named ‘BACKGROUND’ (all caps!) to it!

Try it now and see how the logo and bottom bar disappear!

Presentation tips

Making your slides

- Text boxes are great for presentations
- Remember the shortcuts for new pages, new views, and new layers – you'll be needing them
- Transfer good coding practices to making presentations and make use of pages and layers
- The subheader command or subheadline text style are great for separating topics on a page:
 - Stay in the frame (grid pattern box)!

Presenting your slides

- Your slides can be presented with any PDF viewer
- IPE comes with the IPE presenter application which works great
 - Don't overload a page visually

Conclusion

Further resources

- The official IPE Manual:
<https://otfried.github.io/ipe/index.html>
(or via *Help* > *ipe manual* from inside IPE)
- For making your own presentation template:
https://olejorik.github.io/post/ipe_presentation_template/
- The Github link for this presentation and its README:
missing

Final remark

- IPE is really easy to get a hang of and we hoped this tutorial helped you get a good feel for it!
- We encourage you to take a look at the style file used in this presentation, modify it, or make your own!
- Create your next presentation and your figures with IPE to impress Prof. Storandt!

